

ADDENDUM TO HOMEWORK 3

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1. CONCRETE CONTRIBUTIONS

I created the file `gradient.jl` along with the entirety of its content. The file contains the Armijo algorithm for determining the step size and the general gradient descent function.

2. TECHNICAL DECISIONS

I decided to use Mersenne Twister as my pseudo random number generator for the random initialization due to some online suggestions mentioning its improvements over the generic pseudo random number generator.

I also decided to make the functions as general as possible for future use, such as passing in the algorithm for determining the step.

3. PROBLEMS ENCOUNTERED AND SOLUTIONS

Initially, I had set the maximum iterations to a low number, without letting it converge. After some discussions with my peers, it was evident that I did not have enough iterations. Hence, once letting it iterate enough times, I found a more conclusive result.

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